

Overnight report, 2026-09-13/14: verbose levels, STD strength matching, and seven manuscript improvements

Session 90c5825b on R5611351, branch main, commits f5c5cac .. see §9. Plan: ~/.claude/plans/dreamy-dazzling-ladybug.md. Everything ran through the MATLAB MCP; cClear cClasses was called several times after classdef edits, which WIPED THE BASE WORKSPACE of the live MATLAB session (the variables cfg, cfg3, results, results3, run_dir, run_dir3, zz_*, r_* left in it are mine).

0. The headline: the scaled matching makes the multiple-timescale network chaotic

TR chose (2026-09-13) to match STD strength by a **multiplicative route scale** $s = 1 + r_{\text{ref}}/\rho = 3$ on the two-timescale routes, with the usage-matched (τ_{rel}) variant as a control. Both were built, tested and run at fast (data/stdscaled_fast, data/stdusage_fast; figures in figs/stdscaled_fast, figs/stdusage_fast). The scale variant does what the plan's context warned it would, and more:

	unmatched (data/topk_med, medium, 15 reps)	scaled, $s = 3$ (data/stdscaled_fast, fast, 3 reps)	usage, $\rho_u = 0.342$ (data/stdusage_fast, fast, 3 reps)
lambda_1 near default: none / single / multiple example run $n = 500$, 20 s, seeds [1 2]	+3.47 / +0.35 / -0.112 (105 of 105 negative) +3.35 / +0.43 / -0.11	+3.65 / +0.61 / +2.24 (0 of 21 negative) +3.77 / +0.43 / +1.99	+3.65 / +0.61 / +1.01 (0 of 21 negative) +3.77 / +0.43 / +0.90
tau sweep, multiple-timescale, slowest tau 1 -> 30 s	-0.112 -> -0.043, every rep negative	+2.27 -> +2.27, every rep positive	+0.86 -> +1.14, every rep positive, no trend
memory capacity, total MC: none / single / multiple K needed by top-K for the multiple-timescale condition	0.103 / 0.259 / 0.590 (15 trials, $p = 6e-5$) 15	0.111 / 0.243 / 0.103 (5 trials) 60, D_KY resolved in 0-50% of jobs	0.111 / 0.243 / 0.146 (5 trials) 15-60 (D_KY resolved in 39-100% of jobs)
1-D sweeps: multiple-timescale is the least stable condition on	none of 7	f_E, mu_EE, mu_EI, mu_II (4 of 7)	n, mu_EI, mu_IE, mu_II (above the single-timescale curve on 6 of 7 sweeps; below it only on mu_EE)

With the 3x undepressed recurrent gain the multiple-timescale network is strongly chaotic and its fading memory collapses to the no-adaptation level; the paper's central computational result does not survive the scale direction. The usage control (R3) is less extreme but points the same way: $\lambda_1 +1.0$ near default, +0.9 in the example run, +0.65 to +1.14 across the tau sweep with no trend, memory capacity 0.146. **At equal steady-state depression strength at r_{ref} , the timescale count alone does not produce the stable regime: the unmatched multiple-timescale advantage leaned on the squared (stronger) depression.** That is the honest answer to Codex sec. 1's confound question, and it is what the manuscript must say.

Because both matchings WEAKEN the multiple-timescale network's depression, a third direction was built and launched at 03:15 (sec. 8): keep the multiple-timescale network exactly as published and STRENGTHEN the single-timescale route to the dual's steady state ($\tau_{\text{rel}} 0.25 \rightarrow 0.0625$, ..._dualStdSingleMatched_3cond_mu8p25). If a single-timescale STD of equal strength also gives $\lambda_1 \sim -0.1$ and MC ~ 0.6 , the paper's claim is about depression strength, not timescale count; if it does not, the timescale structure matters at that strength. paper_config still points at the SCALED preset because that was the instruction; **it should not stay there.** Nothing has been run at medium on any matched variant except the usage run launched tonight (data/stdusage_med).

The matching itself is exact at r_{ref} for both variants (fig_STD_strength_matching): $\theta_{\text{ss}}(r_{\text{ref}}) = 0.0833$ for single, scaled and usage; the largest |log ratio| to the single-timescale route over the occupied rate band [0.199, 0.254]

is 1.11 unmatched, 0.146 scaled, 0.034 usage. Usage is also the better match away from r_{ref} . Caveat on record: per-neuron time-averaged rates are strongly bimodal (near 0 or saturated), so “the operating point” is the population mean rate, not a typical neuron’s rate.

0b. Addendum, 03:5x: the third direction keeps the result

The single-matched run (data/stdsinglematched_fast; dual routes as published, single-timescale route strengthened to the dual’s steady state, τ_{rel} 0.0625) had finished its sweeps and memory capacity when this was written (figures still running):

	unmatched (medium, 15 trials)	single-matched (fast, 5 trials)
total MC: none / single / multiple	0.103 / 0.259 / 0.590	0.111 / 0.385 / 0.583
horizon (s)	0.00 / 0.12 / 0.52	0.00 / 0.30 / 0.48
single vs multiple	$p = 1.2e-4$, $d_z = -2.04$	$p = 0.125$ (floor 0.0625 at 5 trials), $d_z = -1.05$
K needed by top-K, multiple-timescale	15	15 (D_KY resolved 67-100%)

So at EQUAL steady-state depression strength the multiple-timescale network still has the larger fading memory (0.58 vs 0.39), while the single-timescale network gains from the stronger depression (0.24 → 0.39). Both mechanisms contribute; the timescale-count claim survives in the form “at matched strength, two depression timescales extend fading memory further than one”, and the effect size is smaller than the unmatched comparison suggested. The multiple-timescale λ_1 for this run is in `figs/stdsinglematched_fast/fig_local_vs_finite_lle/*_table.md` once the figure pass finishes (the sweeps’ $K = 15$ says it is the published stable network). **Recommendation: make ..._dualStdSingleMatched_3cond_mu8p25 the paper preset** – it changes only the single-timescale condition, keeps every multiple-timescale result, and answers the confound honestly – and run it at medium (clone `stdUsage_med_config`). The Methods paragraph in `STD_strength_matching_2026-09-13.md` sec. 4 then needs its matching sentence turned around (the single-timescale release constant shortened to 0.0625 s so its steady-state output equals the two-timescale routes’ at r_{ref}).

1. r_{ref}

$r_{ref} = 0.25$ = the median mean firing rate of the multiple-timescale condition at the default point of the five well-defined 1-D sweeps of data/topk_med (f_E 0.236, level_of_chaos 0.223, μ_{EE} 0.239, μ_{IE} 0.251, n 0.231; with the two negative-mean sweeps resolved correctly the pooled 105 reps give 0.230 [p_5 0.199, p_{95} 0.254]), rounded to 0.05. Single-timescale 0.269, no adaptation 0.483. The joint 64-point sample, far from the default, has 0.154 / 0.159 / 0.250. Full tables and the replacement manuscript Methods paragraph: `docs/notes/STD_strength_matching_2026-09-13.md` (+pdf).

2. What was built (all committed, all tested; §9 lists the commits)

- **Verbose** (f5c5cac): one setting, 'verbose' | 'minimal' | 'near-none', minimal the default, from paper_config through ctx, every stage, ParamSpaceAnalysis2 and both model classes into the parfor workers; the evalc wrappers in the stages are gone; warnings never gated. A serial 2 x 2 sweep prints three lines. test_verbose_levels.
- **STD strength matching** (d942c51): synapse_config.<pre>.<post>.scale (params.W only; obj.W untouched; dynamics, both Jacobians and jacobian_times consistent by construction; the ESN readout check compares it); presets ..._dualStdScaled_3cond_mu8p25 (primary) and ..._dualStdUsage_3cond_mu8p25 (control); golden fixture regenerated (16 presets); fig_STD_strength_matching (in_paper); std-Scaled_fast_config/_run, stdUsage_fast_config/_run; Equations_stability_paper.md subsection, MTS_STD_notes.md addendum, CLAUDE.md. test_route_scale. The manuscript was NOT edited; its “depression is deliberately not normalized” sentence and the replacement are in the note.
- **Tau sweep on both cell types** (b317923): tau_a_EI alias; the stage picks tau_a_EI when the condition’s E and I ladders are identical; tau_levels.md/.mat manifest proves it (R2: equal E and I ladders at all 7 levels). test_tau_a_EI_alias. The sweep folder is now named by its axis.

- **Numerics gate** (c0a2a22): check J (analytic Jacobian vs central finite differences on the reduced net, kink rows excluded); acceptance thresholds fixed in code on 2026-09-14 before any run, saved in `settings.acceptance`; verdict in the `.mat` and `numerics_verdict.md`; 'jacobian' figure variant. `test_numerics_verification_stage` (~13 min).
- **Imbalance eig-heatmap examples** (c0a2a22): `run_eig_heatmap` `cfg.examples` (`mu_EE` x 0.5 / 1 / 1.5, shared seeds, per example and condition eigenvalues, top-K `lambda_1`, mean rate and `B_E`); `fig_eig_heatmap_imbalance` (in-paper); legacy variables unchanged. `test_eig_heatmap_stage`.
- **Figures** (07dc9a6): `fig_example_timeseries` three-column composite with the local rate and the accumulating finite-time `lambda_1` (`plot_local_and_finite_lle`, `SRNNCellTypePairs.routes_identical`); `fig_local_vs_finite_lle`; `fig_lle_vs_rate`; IQR bands and data-driven limits on `fig_sensitivity_medians`; the unclipped density-strip `fig_sfa_EOC_allStd`; E:I weights title overlap.
- **Memory capacity** (this commit): `fig_memory_capacity` redrawn (data-driven cumulative axis, bootstrap bands, paired horizons with the sign-flip `p` and `d_z` read from the saved stats, a table); `run_memory_capacity` writes `<run>_provenance.md` (commit, preset, mode, seeds, trials, protocol, stats, paths). The 30-trial production run: see §8.

3. Numerics verdict (fast, scaled preset; data/stdscaled_fast/numerics_verification/numerics_ver

check	none	single	multiple	note
A noise-free reshoot order slope ≥ 1	2.01 PASS	2.01 PASS	2.01 PASS	Ralston order 2 despite the kinks
B noisy strong-order slope ≥ 1	2.01 PASS	1.98 PASS	1.99 PASS	ratio 4x per halving
L median $ \lambda_{\text{lambda}_1}$ sra1 - ode45 ≤ 0.05	0.108 FAIL	0.169 FAIL	0.722 FAIL	10-s windows, 2 seeds: finite-time scatter
C Benettin - QR ≤ 0.05 , topK - QR ≤ 0.02	0.037 / 0.006 PASS	0.308 / 0.016 FAIL	0.166 / 0.058 FAIL	reduced net $n = 30$, $T = 12$ s
J rel Frobenius, eigenvalue error $\leq 1e-6$	1e-10 / 7e-9 PASS	5e-11 / 5e-10 PASS	6e-11 / 5e-10 PASS	0 kink rows excluded

The gate works as designed: the convergence and Jacobian claims pass; the finite-time LLE agreement claims cannot be judged on fast windows and must be judged at medium (25 seeds, 20-s windows) or production. The manuscript's Jacobian claim now has a saved artifact.

4. Tau sweep on both cell types

`data/topk_med` (unmatched, `tau_a_E` only, 11 levels x 25 reps): every rep negative, `lambda_1` -0.112 at 1 s rising monotonically to -0.043 at 30 s ($-1/\tau = -0.033$): the slowest SFA rung sets the exponent in the stable regime, as the sweep's header predicts. The old figure's "half the distribution positive" described a Benettin-era run. R2 (scaled, `tau_a_EI`): every rep positive, +2.3 at every level – the network is chaotic and the SFA timescale no longer sets `lambda_1`. R3: +0.86 $\rightarrow$ +1.14, every rep positive, no trend_TEXT.

5. Local vs finite-time LLE (fig_local_vs_finite_lle) and LLE vs rate (fig_lle_vs_rate)

`data/topk_med`, near-default set (7 sweeps at the level nearest the default, 105 networks per condition):

condition	lambda_1 median [IQR]	networks with lambda_1 < 0	local rate > 0, share of time	median frac_local_positive	mean excursion (s)
no adaptation	+3.47 [+1.81, +4.33]	4 / 105	83%	0.91	0.70
single-timescale	+0.35 [-0.06, +0.86]	33 / 105	53%	0.54	0.26
multiple- timescale	-0.112 [-0.113, -0.110]	105 / 105	3%	0.02	0.12

So “transient expansion within long-interval stability” (Codex §3’s stronger claim) describes the SINGLE-timescale regime near the boundary (locally expanding half the time, lambda_1 straddling zero), not the multiple-timescale one, which is quietly stable (locally expanding 3% of the time, lambda_1 pinned at -0.11 by the 10-s SFA rung). Keep the stronger idea in the Discussion.

LLE vs mean rate over 1219 networks (joint sample + 1-D sweeps), Spearman rho [bootstrap 95% CI]: no adaptation -0.44 [-0.50, -0.38] (inverted U: quiet +0.3, mid +3.1, saturated -5.7), single -0.31 [-0.36, -0.25], multiple +0.11 [+0.05, +0.16] (flat at -0.11 across the whole rate range). Rate and stability interact through the operating point and are not interchangeable; in the multiple-timescale regime the rate barely predicts lambda_1 at all.

6. Imbalance eig-heatmap examples (R2, scaled preset, fast: reference at n = 500, imbalanced at n = 250)

example	condition	lambda_1	mean rate	B_E
mu_EE x 0.5 (inhibition- dominant)	none / single / multiple	+1.98 / -0.19 / +4.89	0.06 / 0.05 / 0.05	0.42
reference	none / single / multiple	+3.94 / +0.63 / +1.95	0.43 / 0.25 / 0.30	0.50
mu_EE x 1.5 (excitation- dominant)	none / single / multiple	-4.23 / -0.53 / +1.19	0.87 / 0.71 / 0.35	0.55

The sheet (figs/stdscaled_fast/fig_eig_heatmap_imbalance) reads well: the no-adaptation network goes from chaotic to saturated-and-silent as excitation grows, single-timescale stays near zero, and the scaled multiple-timescale network is chaotic in every column (its occupancy spills past Re = 0 in all three rows). Rerun on the chosen preset at medium before it goes in the paper.

7. Transient gain on the scaled preset (fast, for the record)

Operating-point table: lambda_1 +4.0 / +0.44 / +1.95; alpha(J_xx) at state +8.7 / +7.2 / +6.7; alpha(J) +8.7 / +2.6 / +4.0. Adaptation’s feedback still takes the frozen rate from +6.7 to +4.0 in the multiple-timescale regime, but the trajectory itself is at +1.95: the scale has moved the operating point, not removed the mechanism.

8. Memory capacity

Audit of data/topk_med/memory_capacity/MC_sample_hold_20260912_232101_trials15_* (unmatched preset, medium): 15 of 15 paired trials complete for all three conditions, all summaries finite; total MC 0.103 [0.093, 0.114] / 0.259 [0.218, 0.313] / 0.590 [0.515, 0.664]; horizon 0.00 / 0.12 / 0.52 s; exact sign-flip over 32768 patterns: none vs multiple p = 6.1e-5, d_z = -3.32; single vs multiple p = 1.2e-4, d_z = -2.04. fig_memory_capacity resolves exactly that file for run_dir = data/topk_med. Horizons are quantised to 0.3-s holds, so panel (c) is inherently discrete.

R2 (scaled, fast, 5 trials): 0.111 / 0.243 / 0.103, single vs multiple $p = 0.0625$ (the exact floor at 5 trials) with $d_z = +1.66$ in the WRONG direction. R3 (usage): 0.111 / 0.243 / **0.146** (5 trials)_TEXT.

Unattended chain launched at 03:15 in one MATLAB call (diaries in data/): 1. `stdSingleMatched_fast_run` -> `data/stdsinglematched_fast`, `figs/stdsinglematched_fast` (log `data/stdsinglematched_fast_log.txt`; ~1.7 h); 2. `stdUsage_med_run` -> `data/stdusage_med`, `figs/stdusage_med` (medium: 15 reps, 11 levels, 15 MC trials; equal footing with `data/topk_med`; ~3 h); 3. `run_memory_capacity` at PRODUCTION (30 paired trials) on the UNMATCHED paper preset -> `data/topk_med_mc_production` (log `data/mc_production_log.txt`). This replaces the plan's production MC on the scaled preset, whose fast result is already decisive. Plot it with `fig_memory_capacity('run_dir', 'data/topk_med_mc_production', 'out_dir', ...)` (the runner writes into `<output_dir>/memory_capacity`). If MATLAB is still busy in the morning, the diaries say which job it is on; `run_all_paper_analyses` refuses a non-empty run directory, so a crashed job can be restarted after deleting its folder.

9. Commits, figures, what was left undone

Commits tonight, in order: `f5c5cac` verbose; `d942c51` STD matching; `b317923` `tau_a_EI`; `c0a2a22` numerics gate + imbalance examples; `07dc9a6` figures; `9e39cb3` config handles; `a1bb18b` docs; `eb8e56d` `make_all_paper_figures` verbose order (this bug cost R2 one figure pass: the level was read at line 66 and defined at line 119); `35f70bc` memory capacity figure + provenance; `336532f` the third matching direction (`dualStdSingleMatched`) and the medium usage config; the final commit carries this note and the `fig_STD_strength_matching` title fix.

Figures: `figs/stdscaled_fast/<entry>/` (31 of 31 entries succeeded, 17 in-paper, manifest.md there); `figs/stdusage_fast/` (31 of 31, 17 in-paper). Scratch renders from the unmatched medium data are in `figs/scratch/`.

Wrong turns worth knowing: the Bash tool mangles backslashes in heredocs, so every multi-line edit went through perl scripts written with the Write tool (two figure registrations lost their `@` handles that way, `9e39cb3`); test scripts run in the base workspace and clobber loop variables, so tests were run one per call; a `(1, :)` char argument rejects the logical the old callers pass, so the verbose arguments are unconstrained and validated by `verbose_name`; MC's `capture_git_provenance` writes a `working_changes.patch` when the tree is dirty, which it was during the runs (the figure/provenance edits were uncommitted while R2 ran).

Not done: nothing from the plan was skipped. Raised only: the E:I weights sheet's colour scheme and the sensitivity sheets still show the no-adaptation curve dominating the y range (the adapted curves are readable but compressed; a symmetric-log axis would help); `fig_local_vs_finite_lle`'s row-2 x labels overlap between columns; `plot_memory_capacity_combined` still has the fixed [0 10.9] cumulative axis for direct callers; `test_numerics_verification_stage` takes 13 min because the full-size L check has no size override.

10. What TR should do first

1. Read sec. 0. Both weakening matchings (scale, usage) turn the multiple-timescale network chaotic and remove its memory advantage. Look at `data/stdsinglematched_fast` (sec. 8 chain, job 1): does a single-timescale STD of the dual's strength reproduce $\lambda_1 \sim -0.1$ and MC ~ 0.6 ? Read `figs/stdsinglematched_fast/fig_sensitivity_medians`, `fig_memory_capacity` and `fig_local_vs_finite_lle` tables.
2. Decide what the paper claims. Options: (a) the timescale-count claim, supported only if single-matched does NOT reproduce the stable regime; (b) a combined "multiple timescales AND stronger depression" architecture claim, which the unmatched comparison supports and which the Methods paragraph in `STD_strength_matching_2026-09-13.md` sec. 4 must then be rewritten to say; (c) match at a different quantity (e.g. the occupied mean of theta rather than theta at one rate).
3. Move `paper_config` off the scaled preset (one line) to whichever preset the decision names; the medium usage run (`data/stdusage_med`) and the 30-trial production MC on the unmatched preset (`data/topk_med_mc_production`) will be waiting either way.